1. Introduction: Architects vs. Builders

Selamat pagi, Bali! Look around this room for a second. Right now, you are sitting next to the architects of a future that nobody has built yet. Not the builders. **The architects.**

There is a difference between those two words — and the next thirty minutes are about that difference. About who these kids need to become, and about how we, as the adults in their lives, actually help them get there.

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2. The 65% Reality Check

Sixty-five percent. That is the number that gets me out of bed in the morning. Well, that and my kid using my face as a springboard at 5:00 AM.

Sixty-five percent of the jobs these kids will hold haven't been invented yet. Read that again. **We are raising kids for a world that does not exist yet.** So we cannot prepare them with subjects. Subjects describe a world we already know.

The real question stops being *"what should we teach them to do."* The real question becomes: **who do we help them become?**

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3. The Changing Landscape of AI

And here is why this matters today, not tomorrow. This is from Anthropic — one of the most rigorous studies on what AI can actually do right now. Read the line carefully. It is not slowing down.

AI is already writing code as well as your mid-level engineer. By the time the children in this school graduate, "knowing how to code" will mean about as much as "knowing how to type" meant for our generation. Useful, yes. Important, even. **But not the point.**

Parents in the room — if you came here tonight hoping I would tell you that signing your kid up for a coding bootcamp will future-proof their career, I have bad news. The machines have caught up. Your kid is no longer going to be the family's designated IT tech support — the AI is.

So, the human's job has officially changed. And the way we raise our kids has to change with it.

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4. The Core Thesis

So if the machines are doing the work, what is left for the human?

- Deciding what to build.
- Choosing the problem worth solving.
- Telling the machine where to go.

Which is great, because if AI is going to take over the world anyway, I'd at least like my kid to be the one holding the remote control.

That is a different kind of person. Not a coder — coders follow instructions. **An architect. Architects write them.**

The Thesis: We have to stop teaching kids to be the workers, and start teaching them to be the architects. The visionaries who speak to machines to solve human problems. That is the shift. That is the entire game.

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5. The 6 Human Skills

Subjects describe a world we already know. But **skills** travel into any future, including the one we cannot see. These are six of them:

- 🎨 **Creativity**
- 🧠 **Critical thinking**
- 🧩 **Problem solving**
- 🌱 **Resilience**
- 🤝 **Collaboration**
- 🗣️ **Communication**

💡 *Delivery Note:* By resilience, I don’t mean the power of a kid to complain for 20 minutes straight because you cut their sandwich into triangles instead of squares. It is about self-regulating the frustration and pushing through to achieve the objective.

Notice what is not on this list: memorization, syntax, repetition—anything a machine can do faster and better than your kid. What stays is what is inevitably, beautifully human.

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6. The "How" vs. The "What"

But here is the part of the conversation almost nobody finishes. Agreeing that skills matter is only the first half of the answer. The second half is harder. Because skills do not transfer to the kid just by doing the activity.

Two robotics classes with two different methodologies can produce two completely different kids. Same lesson plan. Same robot kit. Same teacher's salary. **Wildly different outcomes.**

So the real question — the question this entire keynote is about — is not *WHAT* we teach. **It is HOW.**

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7. The Vehicle: STEAM Robotics

Now — there are many activities that can carry these skills if you do them right. Theater works. Sports work. Music. Cooking. Gardening. They all carry some of the six. But we chose one in particular: **STEAM robotics.**

So we have the activity. Now — how do we run it so the skills actually transfer?

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8. The Five Shifts

Five shifts. None of them optional. None of them new on their own. And here is the trick: **Each shift is not just a structural change. Each shift directly trains one or two of the skills we care about.**

So when you put all five together, the skills are not taught — they are absorbed through the environment itself. Let me walk you through them, one at a time.

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9. Shift 1: The Tool (TokyMini)

A six-year-old does not need a laptop. A six-year-old needs something they can grab, drop, paint on, glue together — a tool that speaks the language their body already speaks. So we built one. We call it **TokyMini**.

It is screen-free robotics, designed for ages 6 to 9. Because that is the age window where, frankly, your kid does not need more screen time. **They need more world.**

And as they grow, the tool grows with them. Puzzle pieces become circuits. Circuits then work with AI interfaces. The tool always meets the child where they are.

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10. Shift 2: The Teacher & Socratic Dialogue

When we were in school, the important thing was knowing the right answer. Today, the answer is one Google search away. Knowing the right answer is no longer valuable. **What is valuable is asking the right question.**

So our teachers do not give answers. They train kids to ask better questions. That is called **Socratic dialogue**. Two thousand years old. Still works.

🎙️ *Joke / Audience Connection:* "Which means all those years your toddler spent hitting you with that endless loop of 'Why? But why? But why?'... they weren't actually trying to push you to the brink of insanity. Turns out they were just practicing Socratic dialogue twenty years ahead of schedule. You don't have a stubborn child; you have a miniature Greek philosopher in pajamas."

And here is the beautiful part — every time a teacher asks a question instead of giving an answer, two skills get exercised on the kid's side: **Critical thinking** and **communication**. The kid has to think, and then put the thinking into words. Every. Single. Day.

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11. Shift 3: The Room

The old classroom had rows of desks facing the teacher. The new classroom has no rows. It is a **workshop** where every student has their hands on something, all of the time.

When the room changes, two more skills get exercised:

- **Independence:** Because the kid has to make decisions without raising their hand.
- **Collaboration:** Because the kid two feet away is also building something interesting, and you would be a fool not to learn from them.

Parents, you already know this works. The hour you spent building something with your kid taught them more than a whole afternoon of math drills. Now imagine if their school did that, every day, with a real purpose.

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12. Shift 4: The Curriculum (Project-Based Learning)

Not a sequence of lessons. A sequence of projects the kid actually cares about. We call it **project-based learning**.

And this is where the magic happens, because three of our six skills live inside this one shift:

- **Problem solving:** Because the project will break and they have to figure out why.
- **Resilience:** Because the project will break and they have to come back tomorrow and try again.

- **Creativity:** Because it is their project, they care about it, and they want their personality stamped all over it.

No worksheet ever made a kid resilient. A robot that does not work on the seventh try? *That* makes a kid resilient.

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13. Shift 5: Assessment

The way we measure learning has to change too.

- The old way was a **test** that asks *"did you remember?"*
- The new way is a **portfolio** that asks *"what did you build?”*

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14. Human-Centric Education

Why do we believe this is the future of learning? **Because this is human.**

The old education system was designed for the industrial revolution. Stand in line. Sit in rows. Do the same task at the same time as everyone else. We were manufacturing factory workers. And we did it well, for about a hundred years.

But we are not making factory workers anymore. **We are raising humans.** So our methodology has to be human too. The five shifts are not just pedagogy — they are a refusal. A refusal to keep producing identical kids in identical rooms with identical outcomes.

Every child in our makerspace builds something different. Looks different. Thinks different. That is not a bug. **That is the entire point.**

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15. The Tokylabs Blueprint

Here is the full blueprint:

- 🎯 **WHAT we teach:** STEAM robotics, and the six inevitably human skills.
- ⚙️ **HOW we teach it:** The five shifts you just heard.
- 👤 **WHO:** Who is the kid we are trying to raise? That is the most important question of all.

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16. The Four Programs

Here is how a child moves through our ecosystem. Four distinct programs, where each one grows a specific kind of human:

1. Tech & Tinker

- **Format:** After-school.
- **Focus:** Art plus playful technology.
- **The Outcome: The Creator** — the one who is no longer afraid of a blank page.

2. Robot Crafters

- **Format:** Powered by TokyMini.
- **Focus:** 70% soft skills, 30% electronics and coding.
- **The Outcome: The Thinker** — and yes, the Thinker is still a Creator.

3. The Makerspace

- **Format:** Open projects, real prototypes.
- **Focus:** Real frustration, real resilience. We become their mentors—asking, never telling.
- **The Outcome: The Pusher** — the one with self-initiative, who finishes what they start. And the Pusher is still a Thinker, still a Creator.

4. AI Robotics

- **Format:** Advanced tier.
- **Focus:** Organizing ideas with critical thinking and communicating them in the language of machines, using all the computational thinking they have absorbed.
- **The Outcome: The Architect.**

The Progression: Creator $\rightarrow$ Thinker $\rightarrow$ Pusher $\rightarrow$ Architect.

That is the person we believe will navigate the uncertainties of the future, and reach their own definition of success — not ours.

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17. Conclusion

I want to end with the moment that started all of this for me. A child holds up something they made — a robot, a circuit, a working prototype — and their eyes go wide. Not because it is perfect. **Because it is theirs.**

That look — that intellectual joy of *"I can build that"* — is the whole point. That is what hackathons are for. That is what Tokylabs is for. That is what every one of you in this room is for.

Today, in Bali, we are not running a hackathon. We are igniting the fire of future Architects. Let's go light them.

Terima kasih.